



Robotic Systems I

Lecture 8: Optimization-based Control

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Manipulator Equation in Task-Space:

$$\underbrace{\Lambda(\mathbf{q})}_{\text{"Task-Space Inertia Matrix"}} \underbrace{\ddot{\mathbf{v}}}_{\text{"End-effector acceleration"}} + \underbrace{\eta(\mathbf{q}, \mathbf{v})}_{\text{"Coriolis/Gravity Forces"}} = \underbrace{\mathcal{F}}_{\text{"Wrench at end-effector"}}$$

where:

$$\Lambda(\mathbf{q}) = \mathbf{J}(\mathbf{q})^{-T} \mathbf{M}(\mathbf{q}) \mathbf{J}(\mathbf{q})^{-1} = \underbrace{\left(\mathbf{J}(\mathbf{q}) \mathbf{M}^{-1}(\mathbf{q}) \mathbf{J}(\mathbf{q})^T \right)^{-1}}_{\text{This is numerically more stable}}$$

$$\eta(\mathbf{q}, \mathbf{v}) = \mathbf{J}(\mathbf{q})^{-T} \mathbf{C}(\mathbf{q}, \mathbf{J}(\mathbf{q})^{-1} \mathbf{v}) - \Lambda(\mathbf{q}) \dot{\mathbf{J}}(\mathbf{q}) \mathbf{J}(\mathbf{q})^{-1} \mathbf{v}$$

$$\mathcal{F} = \mathbf{J}^{-T} \boldsymbol{\tau}$$

We work with full poses and twists (not just positions)

Desired pose: $\mathbf{T}_d \in \text{SE}(3)$ Current pose: $\mathbf{T} \in \text{SE}(3)$

Pose error (in Lie algebra):

$$\mathbf{e} = \log(\mathbf{T}_d \mathbf{T}^{-1}) \in \mathfrak{se}(3)$$

Twist (spatial velocity):

$$\mathcal{V} \in \mathbb{R}^6, \quad \mathcal{V}_d \in \mathbb{R}^6$$

Velocity error:

$$\mathcal{V}_e = \mathcal{V}_d - \mathcal{V}$$

Goal: Track pose and twist robustly

Task-space wrench:

$$\mathcal{F} = \mathbf{\Lambda}(\mathbf{q}) \left(\underbrace{\dot{\mathcal{V}}_d}_{\text{Feedforward}} + \underbrace{\mathbf{K}_p \mathbf{e} + \mathbf{K}_d \mathcal{V}_e}_{\text{Feedback}} \right) + \boldsymbol{\eta}(\mathbf{q}, \mathcal{V})$$

where:

- $\mathbf{e} \in \mathfrak{se}(3)$ encodes position & orientation error
- $\mathcal{V}_e$ is the twist error

Final control input (what we send to the robot):

$$\boldsymbol{\tau} = \mathbf{J}_w(\mathbf{q})^T \mathcal{F}$$

Full Controller:

$$\boldsymbol{\tau} = \mathbf{J}_w^T \left(\boldsymbol{\Lambda}(\dot{\mathbf{v}}_d + \mathbf{K}_p \mathbf{e} + \mathbf{K}_d \mathcal{V}_e) + \boldsymbol{\eta} \right)$$

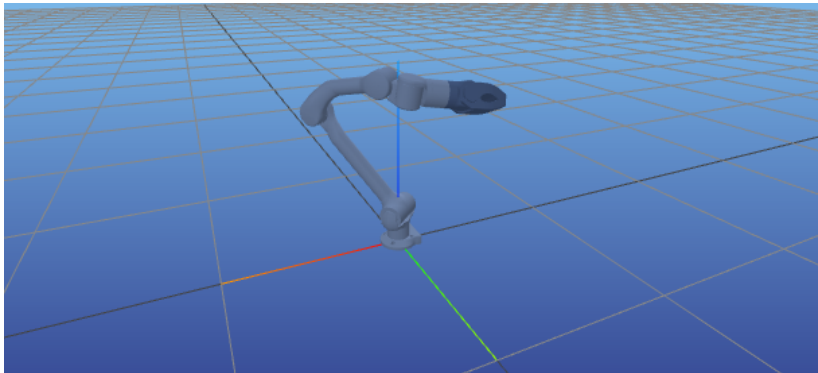
Alternative formulation:

$$\boldsymbol{\tau} = \mathbf{J}_w^T \left(\boldsymbol{\Lambda}(\dot{\mathbf{v}}_d + \mathbf{K}_p \mathbf{e} + \mathbf{K}_d \mathcal{V}_e) \right) + \underbrace{\mathbf{C}}_{\text{Coriolis/gravity}}$$

Important:

- Control is designed in SE(3)
- Actuation happens in joint space

Task-Space Force Control Example



Problem: Redundant robot ($n > m$) $\Rightarrow$ infinite solutions

General solution to differential IK:

$$\mathbf{v}_d = \mathbf{J}^\dagger(\mathbf{q})\mathcal{V}_d + \underbrace{\left(\mathbf{I} - \mathbf{J}^\dagger\mathbf{J}\right)}_{\mathbf{N}(\mathbf{q})}\mathcal{V}_{null}$$

Components:

- $\mathbf{J}^\dagger\dot{\mathbf{x}}_d$: achieves the primary task
- $\mathbf{N}(\mathbf{q})\mathcal{V}_{null}$: motion in the null-space

Typical choice:

$$\mathcal{V}_{null} = \nabla H(\mathbf{q})$$

where $H(\mathbf{q})$ is a secondary objective (e.g., joint limits, posture)

Why is this the Null-Space?

Key property: Null-space motion does *not* affect the task

Apply the Jacobian:

$$J\mathbf{v}_d = JJ^\dagger \mathcal{V}_d + JN\mathcal{V}_{null}$$

Using:

$$N = I - J^\dagger J$$

we get:

$$JN = J - JJ^\dagger J = \mathbf{0}$$

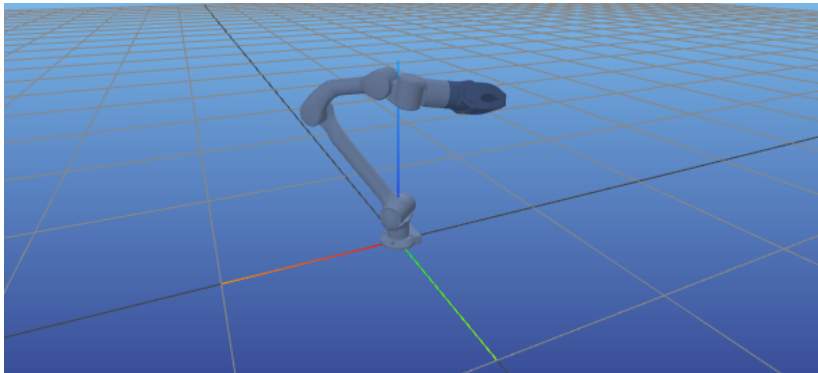
Therefore:

$$J\mathbf{v}_d = \mathcal{V}_d$$

Conclusion:

- Null-space term does not affect the primary task
- It lies in $\ker(J)$

Null-Space Force Control Example



Multiple Task Hierarchies

Goal: Enforce priorities between tasks

Two tasks: $(\mathcal{T}_1, \mathbf{J}_1)$ and $(\mathcal{T}_2, \mathbf{J}_2)$

Hierarchical controller:

$$\mathbf{v}_d = \underbrace{\mathbf{J}_1^\dagger \mathbf{v}_1}_{\text{Primary task}} + \underbrace{\mathbf{N}_1 \mathbf{J}_2^\dagger \mathbf{v}_2}_{\text{Secondary task in null-space}}$$

where:

$$\mathbf{N}_1 = \mathbf{I} - \mathbf{J}_1^\dagger \mathbf{J}_1$$

Extension to more tasks:

$$\mathbf{v}_d = \mathbf{J}_1^\dagger \mathbf{v}_1 + \mathbf{N}_1 \mathbf{J}_2^\dagger \mathbf{v}_2 + \mathbf{N}_1 \mathbf{N}_2 \mathbf{J}_3^\dagger \mathbf{v}_3 + \dots$$

Key idea:

- Each task operates in the null-space of higher-priority tasks
- Lower-priority tasks never interfere with higher-priority ones

Null-Space Control in Torque Control

Key idea: Null-space concepts extend naturally to torque control

Task-space torque (primary task):

$$\tau_1 = \mathbf{J}^T(\mathbf{q})\mathcal{F}$$

Add null-space torque:

$$\tau = \underbrace{\mathbf{J}^T\mathcal{F}}_{\text{Primary task}} + \underbrace{\mathbf{N}^T(\mathbf{q})\tau_{null}}_{\text{Null-space torque}}$$

where:

$$\mathbf{N}(\mathbf{q}) = \mathbf{I} - \mathbf{J}^\dagger \mathbf{J}$$

Property: Null-space torques do not affect task-space motion

$$\mathbf{J}^{-T} \mathbf{N}^T = \mathbf{0}$$

Interpretation:

- τ_{null} : secondary objectives (posture, limits, energy)
- Primary task remains unaffected

Many robots have multiple operational tasks:

- left hand pose
- right hand pose
- left foot pose
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- center of mass
- gaze direction, torso orientation, etc.

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$\mathcal{V}_{i,b}$ = desired body twist / task error

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Goal:

$$J_i(\mathbf{q})\mathbf{v} \approx \mathcal{V}_{i,b} \quad \forall i$$

Multi-Limb IK as a Stacked Least-Squares Problem

Stack all task Jacobians:

$$\underbrace{\begin{bmatrix} \mathcal{V}_1 \\ \mathcal{V}_2 \\ \vdots \\ \mathcal{V}_k \end{bmatrix}}_{\mathcal{V}} = \underbrace{\begin{bmatrix} \mathbf{J}_1 \\ \mathbf{J}_2 \\ \vdots \\ \mathbf{J}_k \end{bmatrix}}_{\mathbf{J}} \mathbf{v}$$

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Stack desired task-space velocities/errors:

$$\mathcal{V}_b = \begin{bmatrix} \mathcal{V}_{1,b} \\ \mathcal{V}_{2,b} \\ \vdots \\ \mathcal{V}_{k,b} \end{bmatrix}$$

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Stack desired task-space velocities/errors:

$$\mathcal{V}_b = \begin{bmatrix} \mathcal{V}_{1,b} \\ \mathcal{V}_{2,b} \\ \vdots \\ \mathcal{V}_{k,b} \end{bmatrix}$$

Then solve:

$$\min_{\mathbf{v}} \|\mathbf{J}\mathbf{v} - \mathcal{V}_b\|^2$$

Optimization variable:

$$\mathbf{x} = \mathbf{v} \in \mathbb{R}^n$$

Multi-Limb IK as a QP

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QP form:

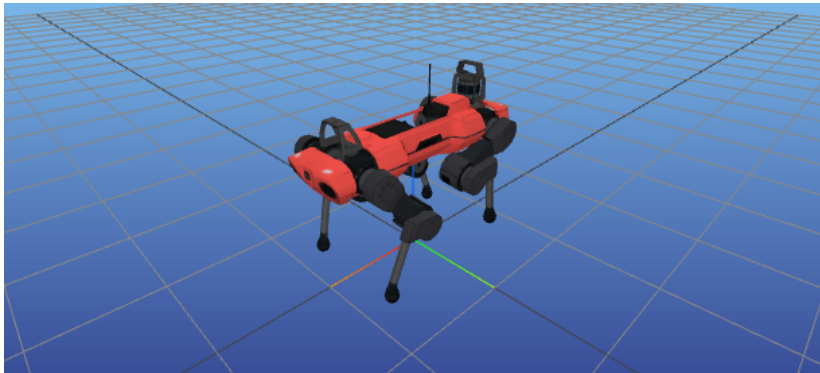
$$\min_{\mathbf{v}} \frac{1}{2} \mathbf{v}^T \mathbf{Q} \mathbf{v} + \mathbf{q}^T \mathbf{v}$$

with:

$$\mathbf{Q} = \mathbf{J}^T \mathbf{J}$$

$$\mathbf{q} = -\mathbf{J}^T \mathcal{V}_b$$

Multi-Limb IK Example



Not all tasks are equally important

Example:

- feet should stay fixed
- hands should move
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Weighted objective:

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- feet should stay fixed
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Weighted objective:

$$\min_v \|W(Jv - \mathcal{V}_b)\|^2$$

QP form:

$$Q = J^T W^T W J$$

$$q = -J^T W^T W \mathcal{V}_b$$

Higher weight $\Rightarrow$ task is tracked more strictly.

Adding Practical Constraints

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Fixed contact constraints:

$$\mathbf{J}_{\text{contact}}(\mathbf{q})\mathbf{v} = \mathbf{0}$$

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Full QP:

$$\min_{\mathbf{v}} \quad \frac{1}{2} \mathbf{v}^T \mathbf{Q} \mathbf{v} + \mathbf{q}^T \mathbf{v} \quad \text{s.t.} \quad \mathbf{A} \mathbf{v} = \mathbf{b}, \quad \mathbf{G} \mathbf{v} \leq \mathbf{h}$$

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- The support region is defined by the contacts (e.g., feet)

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Two regimes:

- **Quasi-static:** slow motions, neglect inertia
- **Dynamic:** fast motions, inertia matters (ZMP, etc.)

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In this lecture:

We focus on quasi-static balancing

Center of Mass (CoM)

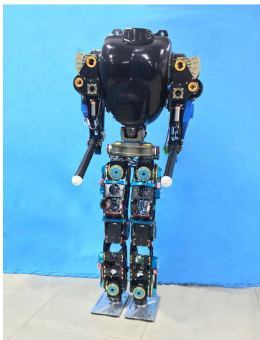
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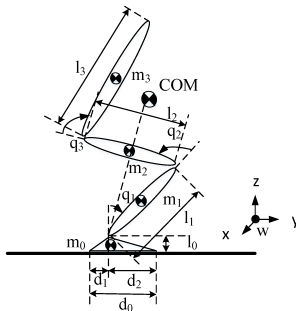
$$\mathbf{x}_{\text{com}}(\mathbf{q}) \in \mathbb{R}^3, \dot{\mathbf{x}}_{\text{com}} = \mathbf{J}_{\text{com}}(\mathbf{q})\mathbf{v}$$

Interpretation:

- CoM summarizes the mass distribution
- Gravity acts as if concentrated at the CoM



(a)



(b)

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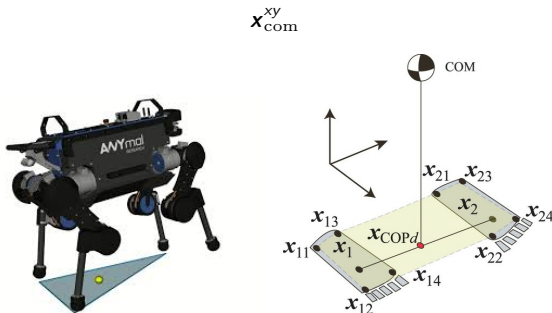
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Projection on the ground:



Key condition:

$$\mathbf{x}_{\text{com}}^{\text{xy}} \in \mathcal{P}$$

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Interpretation:

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Equivalent view:

- All contact forces remain positive (no pulling)

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Desired behavior:

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QP objective term:

$$\left\| \mathbf{J}_{\text{com}}\mathbf{v} - \dot{\mathbf{x}}_{\text{com}}^d \right\|^2$$

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QP formulation:

$$\begin{aligned} \min_{\mathbf{v}} \quad & w_{\text{foot}} \|\mathbf{J}_{\text{foot}} \mathbf{v}\|^2 + w_{\text{com}} \|\mathbf{J}_{\text{com}} \mathbf{v} - \dot{\mathbf{x}}_{\text{com}}^d\|^2 \\ & + w_{\text{swing}} \|\mathbf{J}_{\text{swing}} \mathbf{v} - \mathcal{V}_{\text{swing},b}\|^2 \end{aligned}$$

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Key idea:

- CoM task ensures balance
- Other tasks operate within this constraint

Practical Notes on Balancing

- Often control only $\mathbf{x}_{\text{com}}^{\text{xy}}$ (horizontal plane)
- Height (z) can be controlled separately
- High weight on support contacts $\Rightarrow$ rigid stance
- CoM task must be sufficiently strong to prevent drift

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Limitations of quasi-static approach:

- No dynamic effects (momentum, inertia)
- Cannot handle fast motions (running, jumping)

Balancing Example



Bimanual Velocity Control as a QP controller

Goal: compute joint velocities that track both hands while avoiding self-collisions

Decision variable:

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Velocity commands from pose errors:

$$\mathcal{V}_{L,b} = \mathbf{K}_{p,L} \log\left(\mathbf{T}_L^{-1} \mathbf{T}_{L,d}\right)$$

$$\mathcal{V}_{R,b} = \mathbf{K}_{p,R} \log\left(\mathbf{T}_R^{-1} \mathbf{T}_{R,d}\right)$$

Level 1: Self-Collision Avoidance

Collision avoidance should have highest priority

For two robot points:

$$s(\mathbf{q}) = \|\mathbf{p}_a(\mathbf{q}) - \mathbf{p}_b(\mathbf{q})\|^2$$

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Time derivative:

$$\begin{aligned}\dot{s} &= 2 (\mathbf{p}_a - \mathbf{p}_b)^T (\dot{\mathbf{p}}_a - \dot{\mathbf{p}}_b) \\ &= \underbrace{2 (\mathbf{p}_a - \mathbf{p}_b)^T (\mathbf{J}_{p_a} - \mathbf{J}_{p_b})}_{\mathbf{J}_s(\mathbf{q})} \mathbf{v}\end{aligned}$$

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Safety requirement:

$$s(\mathbf{q}) \geq s_{\text{safe}} = d_{\text{safe}}^2$$

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QP inequality form:

$$-\mathbf{J}_s(\mathbf{q})\mathbf{v} \leq \alpha (s(\mathbf{q}) - s_{\text{safe}})$$

Level 2: Left and Right Hand Tasks

Stack both hand tasks:

$$\mathbf{J}_H = \begin{bmatrix} \mathbf{J}_L \\ \mathbf{J}_R \end{bmatrix}, \quad \mathbf{v}_{H,b} = \begin{bmatrix} \mathbf{v}_{L,b} \\ \mathbf{v}_{R,b} \end{bmatrix}$$

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Hand-tracking QP objective:

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Hand-tracking QP objective:

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But subject to Level 1 safety:

$$-\mathbf{J}_s(\mathbf{q})\mathbf{v} \leq \alpha (s(\mathbf{q}) - s_{\text{safe}})$$

The hands move only if doing so remains collision-safe.

Level 3: Posture and Joint-Limit Avoidance

Use remaining redundancy for comfortable motion

Posture velocity:

$$\mathbf{v}_0 = -\mathbf{K}_q (\mathbf{q} - \mathbf{q}_{\text{ref}})$$

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Subject to higher-priority constraints:

$$-\mathbf{J}_s \mathbf{v} \leq \alpha (\mathbf{s} - \mathbf{s}_{\text{safe}})$$

$$\mathbf{J}_H \mathbf{v} \approx \mathcal{V}_{H,b}$$

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Practical meaning:

- collision avoidance first
- hand tracking second
- posture optimization last

Full Velocity-QP Controller

At every control step, solve:

$$\begin{aligned} \min_{\mathbf{v}} \quad & w_H^2 \|\mathbf{J}_H \mathbf{v} - \mathcal{V}_{H,b}\|^2 + w_q^2 \|\mathbf{v} - \mathbf{v}_0\|^2 \\ \text{s.t.} \quad & -\mathbf{J}_s \mathbf{v} \leq \alpha (\mathbf{s} - \mathbf{s}_{\text{safe}}) \\ & \mathbf{v}_{\min} \leq \mathbf{v} \leq \mathbf{v}_{\max} \end{aligned}$$

At every control step, solve:

$$\begin{aligned} \min_{\mathbf{v}} \quad & w_H^2 \|\mathbf{J}_H \mathbf{v} - \mathcal{V}_{H,b}\|^2 + w_q^2 \|\mathbf{v} - \mathbf{v}_0\|^2 \\ \text{s.t.} \quad & -\mathbf{J}_s \mathbf{v} \leq \alpha (\mathbf{s} - \mathbf{s}_{\text{safe}}) \\ & \mathbf{v}_{\min} \leq \mathbf{v} \leq \mathbf{v}_{\max} \end{aligned}$$

Soft Hierarchy:

- Safety: hard constraint
- Hands: high-weight task
- Posture: low-weight task

At every control step, solve:

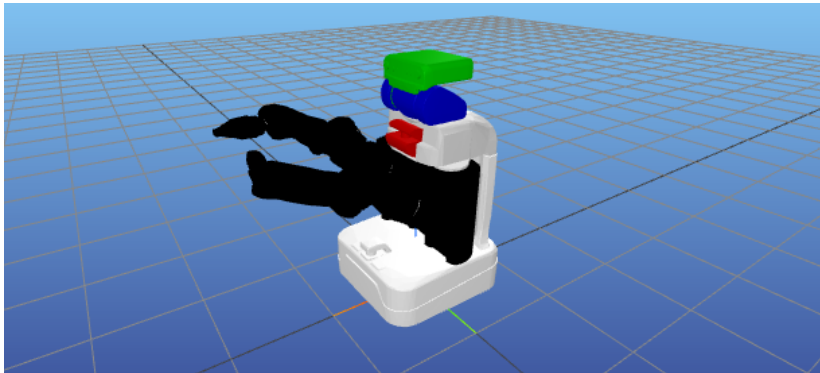
$$\begin{aligned} \min_{\mathbf{v}} \quad & w_H^2 \|\mathbf{J}_H \mathbf{v} - \mathcal{V}_{H,b}\|^2 + w_q^2 \|\mathbf{v} - \mathbf{v}_0\|^2 \\ \text{s.t.} \quad & -\mathbf{J}_s \mathbf{v} \leq \alpha (\mathbf{s} - \mathbf{s}_{\text{safe}}) \\ & \mathbf{v}_{\min} \leq \mathbf{v} \leq \mathbf{v}_{\max} \end{aligned}$$

Soft Hierarchy:

- Safety: hard constraint
- Hands: high-weight task
- Posture: low-weight task

This is still task-space control, but the command sent to the robot is joint velocity.

Bi-manual Control Example



Optimization Based Full Body Control for the Atlas Robot, *Feng, S., Whitman, E., Xinjilefu, X. and Atkeson, C.G.*, 2014, Humanoids. [paper](#)

Thank you

- **Any Questions?**
- **Office Hours:**
 - **Mon-Tue** (09:00-11:00)
 - 24/7 by email (costashatz@upatras.gr, subject: *ECE_RSI_AM*)
- **Material and Announcements**



Laboratory of Automation & Robotics